Lesson 25 Notes:

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1. Matplotlib vs Seaborn :Core Idea

Matplotlib

- Low-level, very flexible.
- Works with arrays/lists.
- You handle most stats and styling manually.

Seaborn

- Built on top of Matplotlib.
- Works directly with **pandas DataFrames**.
- Designed for **statistical plots** (aggregates, CIs, regression).
- Better-looking defaults (colors, grids, styles).

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2. Basic Syntax Difference

```
import matplotlib.pyplot as plt
import seaborn as sns

# Matplotlib
plt.scatter(df['study_hours'], df['exam_score'])
plt.title('Matplotlib')
plt.xlabel('Study Hours')
plt.ylabel('Exam Score')
plt.show()

# Seaborn
sns.scatterplot(data=df, x='study_hours', y='exam_score')
plt.title('Seaborn')
plt.xlabel('Study Hours')
```

```
plt.ylabel('Exam Score')
plt.show()
```

- Matplotlib: pass arrays/Series.
- Seaborn: pass `data=` and column names.

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3. Aesthetics & Defaults

Matplotlib:

```
plt.hist(df['exam_score'], edgecolor='black')
```

Seaborn:

```
sns.histplot(data=df, x='exam_score')
```

- Seaborn automatically applies nicer styles (grid, colors, etc.).

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4. Statistical Aggregation

Matplotlib (manual mean per category):

```
course_means = df.groupby('course')['exam_score'].mean()
plt.bar(course_means.index, course_means.values)
```

Seaborn (automatic mean + CI):

```
sns.barplot(data=df, x='course', y='exam_score')
```

- `barplot` computes mean and shows confidence intervals by default.

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5. Combining Seaborn + Matplotlib

```
sns.boxplot(data=df, x='course', y='exam_score')
plt.title('Exam Scores by Course')
plt.axhline(df['exam_score'].mean(), color='r', linestyle='--')
plt.show()
```

- Seaborn draws main visualization.
- Matplotlib adds titles, lines, annotations.

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6. Example with Built-in Seaborn Data (`fmri`)

Simple line plot:

```
fmri = sns.load_dataset("fmri")

sns.lineplot(data=fmri, x="timepoint", y="signal")
plt.title("fMRI Signal Over Time")
plt.show()
```

Add extra dimensions:

```
sns.lineplot(
    data=fmri,
    x="timepoint", y="signal",
    hue="region", style="event"
)
```

Facet by region:

```
sns.relplot(
    data=fmri,
    x="timepoint", y="signal",
    hue="event", col="region",
    kind="line"
)
```